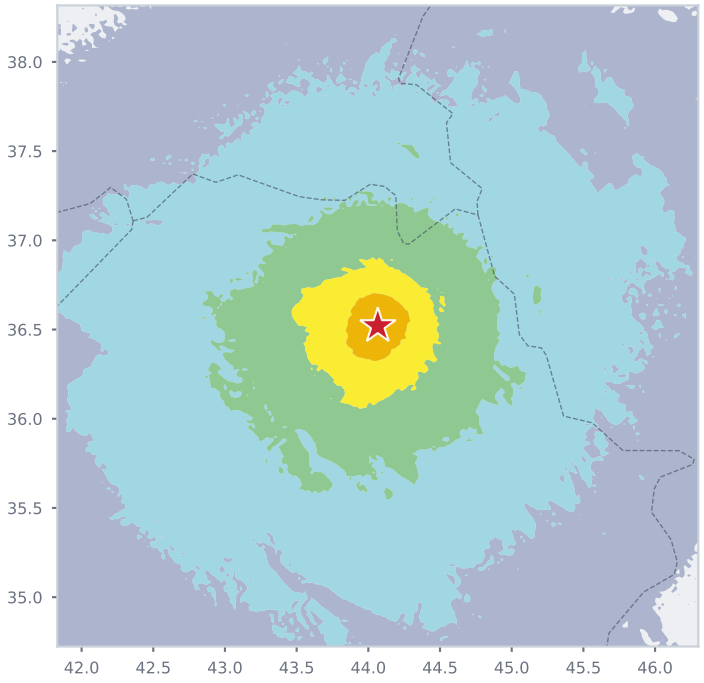
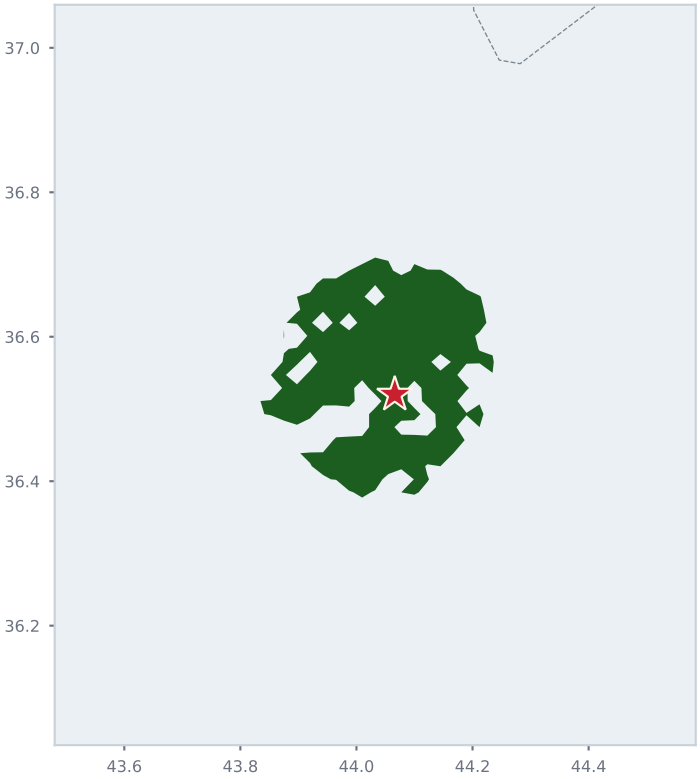
Shaking intensity (IMS-25)



Expected building damage grade



IMS-25 intensity

I II III IV V VI VII VIII IX X XI XII

Expected damage grade (IMS-25)

DG1 DG2 DG3 DG4 DG5

Damage alert

ORANGE Widespread damage; many buildings unusable

Provisional **Day occupancy**

Buildings heavily damaged (DG3 or worse)

1.7 thousand to 38 thousand

Central estimate 7.6 thousand. The range is the 5th to 95th percentile.

Buildings in the shaken area

3.4 million

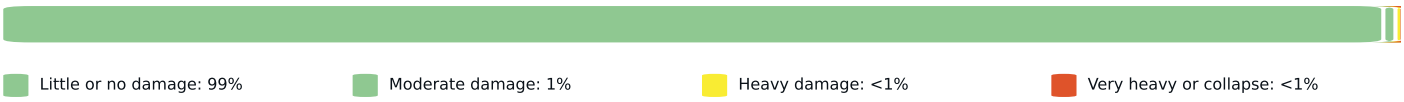
21% of them inferred from built-up area, not observed footprints.

People by shaking level

VIII	2.6 thousand
VII	75 thousand
VI	2.4 million
V	3.8 million
IV	5.7 million

Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage



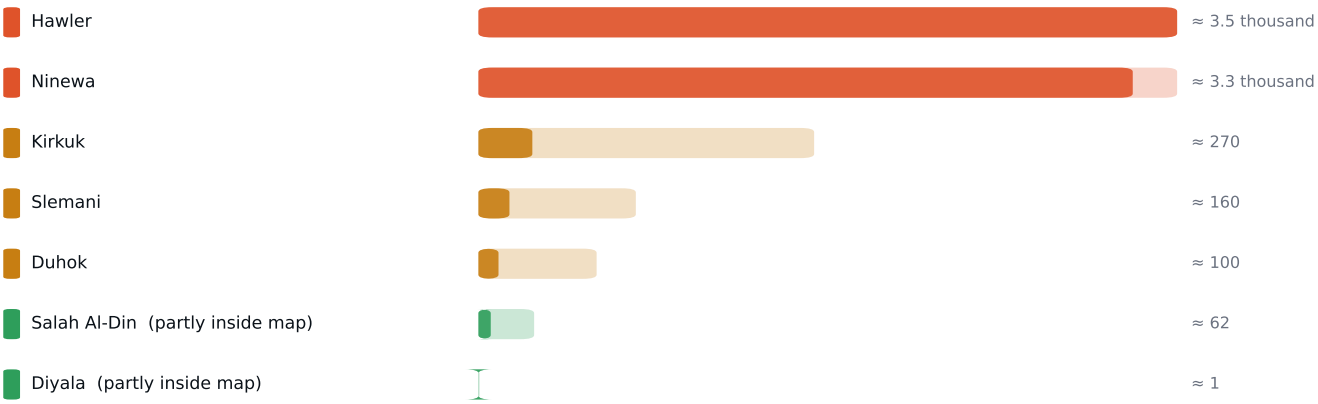
Little or no damage: 99% | Moderate damage: 1% | Heavy damage: <1% | Very heavy or collapse: <1%

Which buildings failed

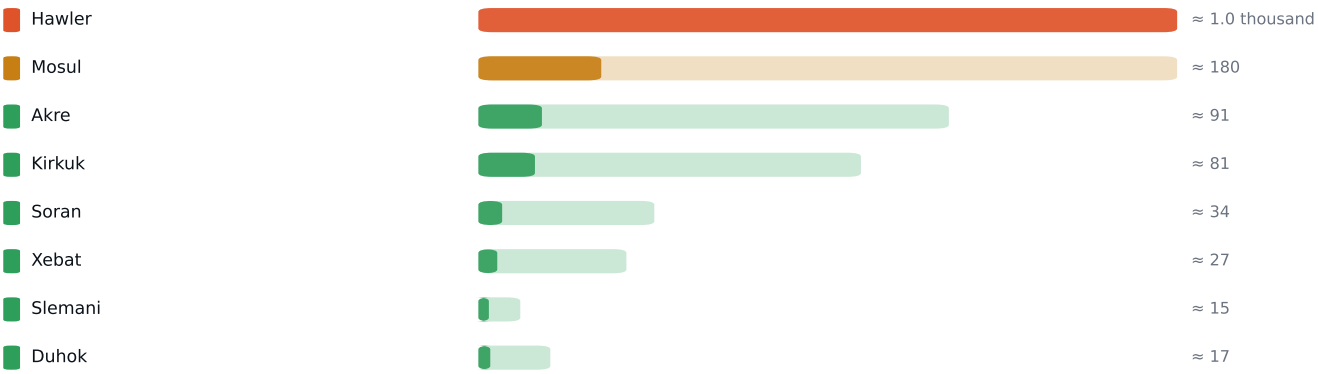
Share of each type's own buildings heavily damaged, worst first

Rubble stone, fieldstone	class A	**1.1%**	≈ 1.1 thousand buildings
Adobe, earth brick	class A	**0.8%**	≈ 830 buildings
Manufactured stone units, timber floors	class B	**0.2%**	≈ 970 buildings
Simple stone	class B	**0.2%**	≈ 620 buildings
RC flat slab and others	class B	**0.1%**	≈ 4 buildings
Manufactured stone with RC floors	class C	**0.1%**	≈ 1.1 thousand buildings
RC frame — no / low code (ERD-L)	class C	**0.1%**	≈ 140 buildings

Governorates most affected: heavily damaged buildings, central estimate and 90% range



Cities most affected



Every governorate inside the map

Area	Buildings	Heavy (P50)	90% range	% of its stock
Hawler	659 thousand	2.4 thousand	790 to 20 thousand	0.36%
Ninewa	960 thousand	2.1 thousand	670 to 16 thousand	0.22%
Kirkuk	570 thousand	200	73 to 1.7 thousand	0.03%
Slemani	632 thousand	110	40 to 790	0.02%
Duhok	266 thousand	69	26 to 590	0.03%
Salah Al-Din	261 thousand	46	18 to 280	0.02%
Diyala	11 thousand	1	none to 6	0.01%

How this report was made

Hazard engine Bumelerze SHAKEmap 26.2.0, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)

Observations 0 seismic stations, 0 felt-report cells (felt channel: none)

Damage scale IMS-25

Data sources

Built from USGS, EMSC and GEOFON earthquake parameters; building footprints from GFZ OpenBuildingMap (c) OpenStreetMap contributors, ODbL, and Microsoft, CDLA-Permissive-2.0; JRC Global Human Settlement Layer and WorldPop, CC BY 4.0; OCHA COD-AB Iraq, CC BY-IGO; GEM Global Exposure Model; IMS-25 vulnerability table. Full records, with licences and checksums: github.com/Peshawa-LH/bumelerze-engine

What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Hasan P. L. (2026) Bumelerze SHAKEmap and damage report, event bml19910001, version 2, generated 19 Oct 2026, 12:41 UTC.